

28th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2024)

Specification description analysis method for requirement modeling

Masami HAYASHI^{a*}, Kaori HAYASHI^a, Yohei SONOBE^a,
Hiroyuki SATO^a and Hironori TAKEUCHI^b

^aDENSO CREATE INC. 2-14-19 Nishiki, Naka-ku Nagoya-shi, Aichi, 460-0003 Japan

^bMusashi University 1-26-1 Toyotamakami, Nerima-ku, Tokyo, 176-8354 Japan

Abstract

In the development of software systems, multiple members create models of requirement specifications under the guidance of experienced professionals. In such cases, the time taken to create models and the quality of the resulting models depends on the members' system development experience and the characteristics of the specifications. Consequently, the cost of modeling and the quality of deliverables can make development more challenging. In this study, we identified the elements necessary to develop models of uniform quality during the process of converting specifications into the requirements analysis model defined in this study. We developed a modeling method that consists of information that should be agreed upon in advance between experienced professionals and members, and patterns of specification description analysis.

© 2024 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the 28th International Conference on Knowledge Based and Intelligent information and Engineering Systems

Keywords: Requirements Specifications; Requirements Analysis; Software Development; Modeling;

1. Introduction

In software development, it is necessary to understand the requirement specifications of the development target before proceeding with the design and production processes. Modeling is effective for understanding requirement

* Corresponding author. Tel.: +81-52-728-0771;

E-mail address: masami.hayashi.j4p@jpgr.denso.com

specifications[1]. For instance, it has been reported that by using the Unified Modeling Language (UML), such as a Data Flow Diagram (DFD)[2] for modeling, it is possible to visually detect omissions in the requirements specification. It is common to model content to understand the requirement specifications in this manner [3]. When developing large-scale software for a team, there are situations in which experienced professionals and practitioners with different skills or experience model the requirement specifications for a single system.

However, when modeling in team-based software development, if a member with little development experience uses notations such as UML, they must first understand the notation. Additionally, there may be a lack of consistency and uniformity between the models created by professionals and those with less development experience, leading to models that are not of sufficient quality and require additional effort for adjustments, such as consistency. Consequently, a problem arises where modeling incurs significant costs.

This study developed a modeling process for team-based software development and specific modeling techniques (hereafter referred to as modeling techniques) to address the quality issues that arise in modeling by members with different levels of development experience. The effectiveness and challenges of the developed process and modeling techniques were confirmed through practice. Although the types of models dealt with in modeling work are diverse, it is necessary to limit the models to specific process evaluations and element assessments. Therefore, the models addressed in this study are based on the requirements analysis model utilized by a company affiliated with the authors.

The remainder of this paper is organized as follows. In Section 2, related studies are described. In Section 3, we introduce the requirements analysis model used in this study and define the research challenges. In Section 4, we describe the proposed business analysis method. In Section 5, we describe the practice to confirm the effectiveness of the proposed method. Finally, we discuss the results in Section 6, and summarize the key points and future studies in Section 7.

2. Related work

Research has been conducted on modeling requirement specifications to eliminate errors and omissions in specifications written in natural language. Ohnishi proposed a method of creating multiple models using UML from requirement specifications that removed errors such as omissions and ambiguities and corrected inconsistencies by verifying the consistency between models[4]. Aoki modeled the requirement specifications for object-oriented system development methodologies based on an Object Modeling Technique and proposed a consistency verification method focused on data flow[5].

On the other hand, research has been conducted to establish the writing style and format of requirement specifications in the process of modeling them. Marvin proposed a simplified syntax called the Easy Approach to Requirements Syntax (EARS) to describe requirements in natural language, advancing the resolution of problems such as ambiguities in requirement definitions[6]. Miyazaki developed an EARS for Japanese requirement specifications, showing that it can reduce ambiguity in Japanese requirement specifications[7]. Additionally, Ohnishi developed a requirement description language called X-JRDL, proposing requirement specifications that use demonstrative pronouns, phrase omissions, compound sentences, and complex sentences[8][9][10]. Furthermore, Cin proposed rewriting specifications in a notation called SQIRL to understand specifications that include nonfunctional requirements[11].

3. Research Subjects and Challenges

In this section, we describe the requirements analysis model employed as a foundational element in this study, as well as the targets for model conversion. Furthermore, we elaborate on the challenges this study addresses during the process of converting requirement specification descriptions into the requirements analysis model.

3.1. Requirements analysis model

The requirements analysis model is a structured model that organizes the necessary elements and their relationships to achieve a common understanding among stakeholders regarding software development projects[12]. Our company adopted this model to facilitate the understanding of requirements. For example, the created requirements analysis

model is cross-referenced with past accumulated knowledge to identify deficiencies in specifications and points of attention in the design process.

The elements of the model and their definitions are listed in Table 1. The requirement specification document contains multiple descriptions of the desired functionalities and behaviors that constitute them. In the requirements analysis model, requirement specification descriptions were used to extract and model the elements listed in Table 1.

Table 1. Elements of the requirements analysis model.

Element Name	Definition	Notation
Function Element	Functions that constitute the system. Each has one or more behaviors.	
Data Element	Information that flows between functions. Information used by a function is input data, and information created by a function is output data.	
Trigger Element	The trigger for executing a function itself. A trigger for starting the function itself is an input trigger, and a trigger for starting another function is an output trigger.	

Table 2. Elements of EARS.

Element Name	Definition
pre-conditions	Conditions under which the behavior is activated.
trigger	Event that initiates the behavior.
system name	Name of the system triggered by the trigger.
system response	The behavior of the system caused by the trigger.

The notations for requirement specification descriptions defined in EARS[6] include the elements listed in Table 2. In the EARS, a trigger is defined as the initiation of a behavior, whereas in the requirements analysis model, a trigger indicates the initiation of a function. Meanwhile, in the requirements analysis model, the opportunity to start a function for individual behaviors is considered as the data observed during the realization of a function. This is represented as the data input to the function. For example, if there is a requirement specification description for a function named "Function A" that states "Function A operates when the power source is turned ON. If the value of Data A is 1, then output Data B." The corresponding requirements analysis model is shown in Figure 1.

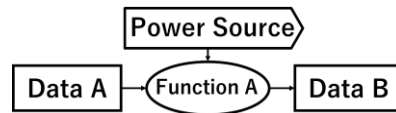


Fig. 1. Example of requirements analysis model

3.2. Description of the Requirement Specifications for Modeling

The descriptions that this study aims to model are those created in the process shown on the right-hand side of Figure 2. This is because the components of the requirements organization model, as shown in Table 1, become clear during the process of creating requirements or specifications that technically define customer demands.

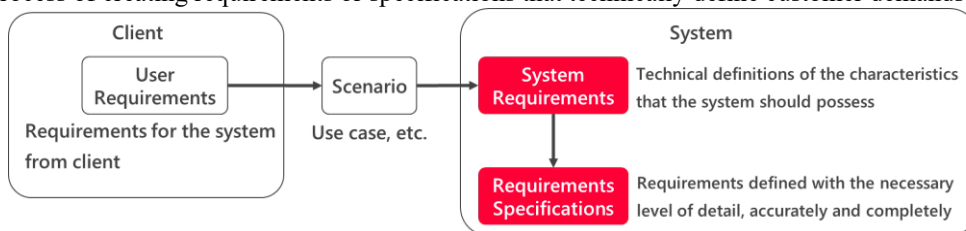


Fig. 2. Process for developing requirements specifications for the target system.

3.3. Challenge

In this study, the requirement specifications described in Section 3.2, are converted into the requirements analysis model presented in Section 3.1. However, when individuals with less experience undertake the task of converting to the requirements analysis model, it is challenging to create models of sufficient quality compared with those produced by experienced professionals. Therefore, in Section 4, we propose a modeling process and methods that can address this challenge and confirm the effectiveness of the proposed approach through a simple experiment in Section 5.

4. Proposal

For the requirement analysis process in this study, we define the following subprocesses by which the project members develop the models under the guidance of experienced professionals:

1. Pre-agreed Process
2. Modeling Process that uses Superficial Information in Specifications
3. Modeling Process that uses Semantic Information in the Texts

To carry out these processes, the necessary information and modeling methods are required. Figure 3 shows the correspondence between each process and the necessary elements. Experienced professionals provide necessary information to other members. This is the information that should be shared among all members before starting the modeling process, which we refer to as “PI”. Then, the members model using the information provided and the modeling method that uses superficial information in specifications, we call this method “MMSIS”. Finally, experienced professionals model descriptions that are difficult for the members to model, based on the modeling method that uses semantic information in the texts using the models created by the members, we call this method “MMSIT”. In this study, we focus on the first and second processes described above and propose the PI and MMSIS in Section 4.1 and 4.2.

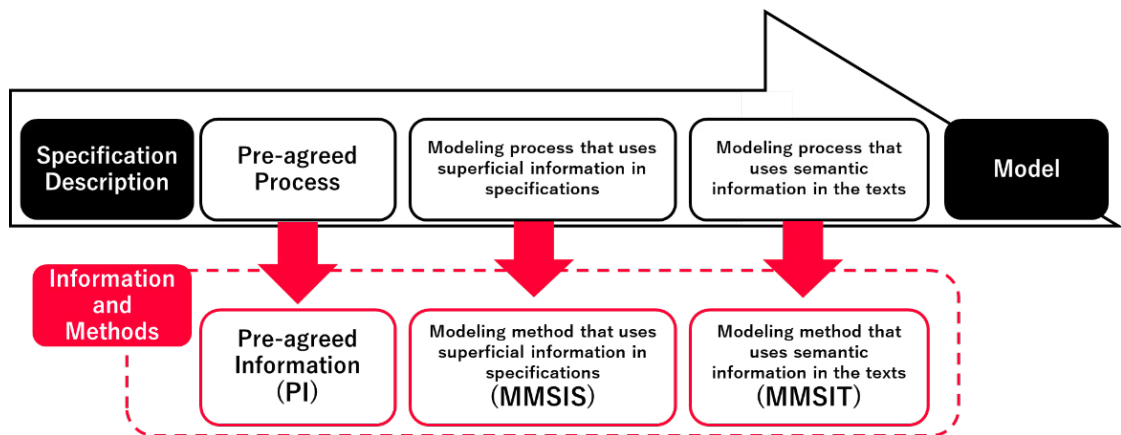


Fig. 3. Information and methods used in the requirement analysis processes.

4.1. Pre-agreed Information (PI)

The PI is the information that must be determined for each specification before performing the Modeling Process that uses Superficial Information in Specifications. Experienced professionals within the team initially present the PI to members, who then use this information to create models. The PI can be broadly categorized into the following types, with specific details discussed later: We identified these types of information based on the principles of requirement modeling[13].

- Information related to the understanding of the document structure of the specification
- Information concerning the identification of the sections to be modeled
- Information regarding the understanding of the notation and descriptions in the specification

4.1.1. Information related to the understanding of the document structure of the specification

Information related to understanding the document structure of the specification does not pertain to specific descriptions within the specification, but rather to information about the overall structure of the specification. This includes the following.

- Order in which to read the specifications: This provides information on the sequence in which to read the chapters and sections of the specifications. For example, if there is a section in the specification in which the data names or terms used in the system are defined, this section should be read first. This enables members to understand the data names and meaning of the text in the requirement specification descriptions that they read later, and to extract these as targets for modeling.
- Granularity of functions: This refers to the level of detail in the specification descriptions corresponding to a single function in the requirements analysis model. As the perception of functional granularity may vary among members, it is necessary to agree on this before modeling based on superficial judgment. In addition, because the unit of function may differ depending on the intended use of the model, experienced professionals must present this information to the members of each project.

4.1.2. Information concerning the identification of the sections to be modeled

The information related to identifying the sections to be modeled indicates the scope within the specification that needs to be modeled and the areas that do not require modeling. For example, sections of the specification that describe functions or behaviors are targeted for modeling and these sections should be indicated. Additionally, supplementary descriptions and diagrams that facilitate the understanding of the descriptions targeted for modeling do not need to be represented as models; therefore, it should be made clear that these do not need to be modeled by members.

4.1.3. Information regarding the understanding of the notation and descriptions in the specification

Information related to the understanding of the specification's notation and descriptions concerns the descriptions and notations that affect modeling and handling. This information includes the following.:

- Use of parentheses: This refers to the use of parentheses within the specifications, including information about their purpose. The definitions of the use of parentheses are based on the classifications proposed by Sugimoto et al.[14], as shown in Table 3. The purpose of parentheses varies by specification, which affects whether the content within them is targeted for modeling. Therefore, experienced professionals must present the use of parentheses for each specification to the members.

Table 3. Use of parentheses.

Type of use	Example
Supplementary Explanation	Output Data A (set the state to B).
Paraphrase	Heat with a heater (keep the heater On).
Meaning Attribution	Data A (Input).
Value, Unit, Example	Percentage (e.g., 100%, 85%).
Reference	System State Value (see section 3.9).

- Dictionary for terminology in the target domain: This guides the handling of model elements for words specific to the field that are not defined within the specification. For members with less development experience, it can be difficult to decide whether an element should be treated as a data or a trigger element without understanding the descriptions. This information must be presented beforehand to facilitate the modeling of these elements.

- Scope of the system for modeling: This determines how to handle requirement specification descriptions that include the actions of a separate system or humans apart from the system being modeled. Similar to "Granularity of functions," the scope of the system to be modeled may vary depending on the intended use of the model. However, for members with less development experience, it can be challenging to model the intended use of the model; thus, this information must be provided explicitly.

4.2. Modeling method that uses superficial information in specifications (MMSIS)

The MMSIS is a modeling method that can process requirement specification descriptions even without extensive experience or knowledge. Using this method, members can create models of consistent quality with the PI provided in advance. In this study, as an MMSIS, we propose five cases: "Sentences containing conditions," "Sentences without conditions," "Sentences listing conditions," "Sentences containing mathematical expressions," and "Sentences where elements of the requirements analysis model are directly written." In the following sections, we explain the modeling approaches for each of these cases.

4.2.1. Sentences containing conditions

Figure 4 illustrates the structure of the sentences containing conditional clauses. For this type of sentence, we extracted words as model elements and constructed a requirements analysis model using the following steps:

- The subject or object corresponding to the predicate of a sentence containing the conditions was extracted as the input data for the function.
- The subject or object corresponding to the predicate of a sentence containing a function's behavior was extracted as the output data for the function.

4.2.2. Sentences without conditions

Figure 5 shows the structure of the sentences without conditional clauses. For this type of sentence, we extracted words as model elements and constructed a requirements analysis model using the following steps.

- The subject or object corresponding to the predicate was extracted as the output data for the function.

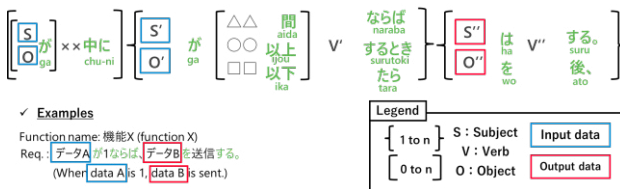


Fig. 4. Method for sentences containing conditions.

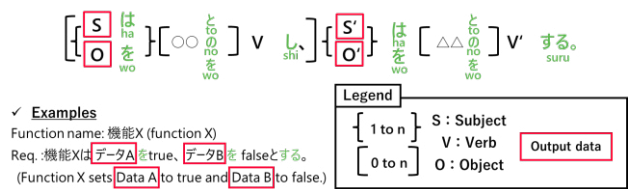


Fig. 5. Method for sentences without conditions

4.2.3. Sentences listing conditions

Figure 6 shows the structure of the sentence-listing condition clauses. For this type of sentence, we extracted words as model elements and constructed a requirements analysis model using the following steps.

- The subject or object corresponding to the predicate of a sentence containing the conditions was extracted as the input data for the function.
- The subject or object corresponding to the predicate of a sentence containing the behavior of a function was extracted as the output data for the function.
- The subject or object of the listed conditions was extracted as the input data for the function.

4.2.4. Sentences containing mathematical expressions

Figure 7 shows the structure of the sentences containing mathematical expressions. For this type of sentence, we extracted words as model elements and constructed a requirements analysis model using the following steps.

- The subject or object corresponding to the terms of the mathematical expression was extracted as the input data for the function.
- The subject or object corresponding to the predicate of a sentence containing the behavior of a function was extracted as the output data for the function.

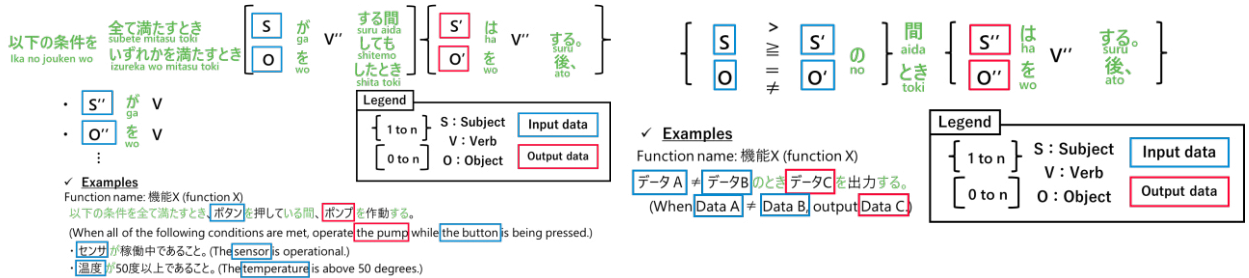


Fig. 6. Method for sentences listing conditions.

Fig. 7. Method for sentences containing mathematical expressions.

4.2.5. Sentences where elements of the requirements analysis model are directly written

Below is the modeling method for sentences within the text that includes elements of the requirements analysis model. In this method, if words in the requirement specification descriptions are directly designated as corresponding to any of the components of the requirements analysis model, they are extracted as is. For example, if there is a requirement specification description stating, "Function A takes Data X as input data," Data X would be extracted as input data.

5. Practice and Results

To verify the effectiveness of the proposed modeling process, we conducted a comparison between models created according to the proposed process and the correct models. In this section, we describe this practice and its results.

The participants, procedures, and target specifications used in the practice are described below.

- Participants
 - Member: An employee with less than one year of development experience in the company to which the author belongs.
 - Experienced Professionals: The author (with one year of development experience).
- Procedure
 1. As an experienced professional, the author presents the PI corresponding to the specifications to be modeled by the member.
 2. The participant models using the PI and the MMSIS.

The requirement specification used for this modeling practice was the software requirement specification for the robot competition. The ease of modeling this specification document using the proposed method was not recognized. The correct model was created separately by an experienced professional using PI and MMSIS. Table 4 lists the PI used in practice.

Table 4. PI used in the practice.

PI	Content	Intent/Reason
Order in which to read the specification	Definitions of terms → Overview of the software → Requirement specifications in this order	To enable the members to grasp the terms and understand the specifications from the outset
Granularity of functions	Each section within the chapter where the requirement specifications are written is considered one function	Because the section names are titled "XX Function" and they contain requirements for the functions
Scope of modeling/ non-modeling	The chapters where the requirement specifications are written are the scope Graphs and diagrams are not included	Because they contain the requirement specifications for the functions They are more like supplements to the requirements
Usage of parentheses	The contents within parentheses are not modeled	Descriptions within parentheses are for supplementary/reference information
Terminology dictionary of the target domain	The word "power supply" is considered as a trigger element	Because "power supply" was managing the operation of functions in the specification descriptions
Scope of the system to be modeled	Specifications that include human involvement are also to be modeled	Because there were requirement specifications in the document that included human involvement

5.1. Results

The requirements analysis models to be created were one model per function requirement, with a total of 10 models. The elements of the models created by the participant and those of the correct models were compared for each function, and the results of the precision and recall calculations are listed in Table 5. In this table, the hyphen (-) indicates that there are no trigger elements in the correct model.

Table 5. Precision and recall of the created model elements.

Function Req. No.	Data element		Function element		Trigger element	
	Precision	Recall	Precision	Recall	Precision	Recall
1	1.00	1.00	1.00	1.00	-	-
2	0.50	0.20	1.00	1.00	-	-
3	1.00	0.33	1.00	1.00	-	-
4	0.89	0.72	1.00	1.00	1.00	1.00
5	1.00	0.83	1.00	1.00	-	-
6	0.40	0.44	1.00	1.00	-	-
7	0.50	0.33	1.00	1.00	-	-
8	0.75	0.60	1.00	1.00	-	-
9	0.63	0.42	1.00	1.00	1.00	1.00
10	0	0	1.00	1.00	-	-
Average	0.67	0.49	1.00	1.00	1.00	1.00

From these results, it was found that the function and trigger elements extracted from the specification descriptions matched the correct models and models created by the participant. Furthermore, it was observed that the participant could extract a fair number of correct data elements; however, there were cases in which not all data elements could be extracted.

6. Discussion

The results showed that the functions and trigger elements of the models created by the participant matched those of the correct models. However, discrepancies were observed in the data elements when compared to the correct models. Based on the results of the practice, we discuss the effectiveness of the proposed methods and the factors contributing to errors in extracting data elements from the models.

6.1. The effectiveness of the proposed methods

The methods proposed in this study aim to provide a solution to the challenge of achieving sufficient model creation quality by members with little development experience. If the proposed methods enable members to create the requirements analysis models of consistent quality, the workload of additional revisions by experienced professionals can be reduced. Therefore, if the quality of the models created by the members becomes consistent through the proposed methods, they can be considered effective. Although it is necessary to verify the quality of the models created by members through further validation with a larger number of participants, the modeling process and methods proposed in this study have the potential to be effective in understanding the requirement specifications in team development.

Because there was only one member involved in this practice, it was difficult to evaluate its effectiveness. However, the results of the practice showed that the extraction of data elements for the models created using the current modeling process and methods did not meet expectations. Because this concerns the effectiveness of the proposed methods, the next section discusses the factors involved.

6.2. Reasons for the discrepancy in the Data elements

There were two main reasons why the participant was unable to correctly extract data elements from the requirement specifications.

The first reason is thought to be the differences in perceptions arising from how the modeling process and methods were explained to the participant. An example of a problematic explanation is the one regarding the "Scope of the System to be Modeled." In this explanation, it was stated that "specifications that include human involvement are also to be modeled." The intention was to convey that the requirement specifications describing human involvement should also be considered in the modeling. However, the participant interpreted the term "user" itself as part of the system to be modeled. An example of a model resulting from this misunderstanding is shown in Figure 8. The requirement specification description modeled in the example was "The switch detection function will 'shortly beep' the buzzer when the user presses the switch.". This shows that the participant extracted data elements that did not match the correct model because of the manner in which the modeling process and methods were explained.

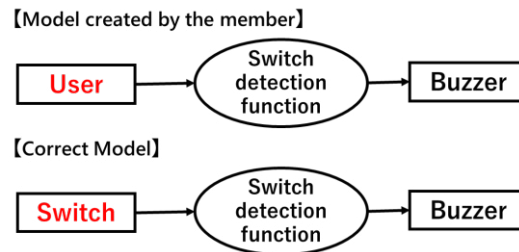


Fig. 8. Example of a model created due to the member's misunderstanding.

The second reason is due to the way the requirement specifications were written. An example of a problematic requirement specification description is "The user places the robot on the course so that the color sensor is directly above the relevant color and collects the reference value of the relevant color by pressing the touch sensor." The latter part of this description indicates the behavior of the system, whereas the former describes the user's actions. Sentences that solely describe the actions of the user should not be considered for modeling because they do not describe the behavior of the system. However, elements of the first part of the description were also reflected in the model created by the participant. This means that, for the participant, it might be difficult to discern whether such descriptions should be modeled. Thus, it can be said that the models were created from sentences that should not have been modeled because of the way the requirement specifications were written.

Regarding the explanation of the modeling process and methods, one approach could be to provide examples of how the modeling techniques are applied. Additionally, formalizing the manner in which requirement specifications

are written could be a solution to the writing style of requirement specifications. For instance, if the requirement specifications are written using the syntax demonstrated in the MMSIS, modeling can be performed more easily. However, in real specifications, there are some cases that we need a deep analysis using the semantic information in the texts and we need MMSIT for such cases. Through experiments by various practitioners, we need to identify the limitations of MMSIS and develop MMSIT. These are our future work.

7. Conclusion

In this study, we proposed a modeling process and methods for multiple members to convert the contents of requirement specifications into requirements analysis models under the guidance of experienced professionals. Specifically, we introduced a process that involves determining the information that experienced professionals and members need to agree upon before modeling and a method that allows members to model based solely on superficial information, enabling them to model independently.

This study aimed to address some future challenges. The first challenge is to innovate the way in which methods are explained to members based on the insights gained from this practice. The second challenge is to rewrite the requirement specification descriptions into a format that is easier to model and implement in practice.

References

- [1] Nuseibeh, B. Kramer, J. and Finkelstein, A. "A framework for expressing the relationships between multiple views in requirements specification," in *IEEE Transactions on Software Engineering*, vol. 20, no. 10, pp. 760-773
- [2] Deeptimahanti, Deva Kumar and Babar, Muhammad Ali. (2009) "An Automated Tool for Generating UML Models from Natural Language Requirements," *2009 IEEE/ACM International Conference on Automated Software Engineering, Auckland, New Zealand*, pp. 680-682
- [3] Wada, T., Yumikawa, Y., Sumi, T., Fujimoto, H., Murata, Y. (2013) "Construction of Traceability and Detecting Deciciencies of Conditional Sentences for Quality Improvement of Requirements Specifications Using Data Flow Diagrams", *Proceedings of the 30th Annual Conference of the Japan Society for Software Science and Technology*: 625-632, (in Japanese)
- [4] Ohnishi, A. (2001) "A supporting system for verification among models of the UML.", *Systems and Computers in Japan*, Volume 33 No.4: 1-13.
- [5] Aoki, T., Katayama, T. (1998) "Formal model approach for reliable Object-Oriented information system design", *Proceedings of the 4th International Conference on Information Systems, Analysis, and Synthesis (ISAS 98)*, pp228-235.
- [6] Marvin, A., Wilkinson, P., Harwood, A., Novak, M. (2009) "EARS: Easy Approach to Requirements Syntax", *RE2009*: 317-322.
- [7] Miyazaki, T., Kitano, S., Saeki, M. (2022) "Developing Japanese Version of EARS to Eliminate Ambiguity in Requirements Specification Documents", *IEICE Technical Report SS2022-03*: 97-102, (in Japanese)
- [8] Onishi, A. (1996) "Software Requirements Specification Database based on Requirements Frame Model.", *Proc. IEEE International Conference on Requirements Engineering (ICRE '96)*: pp221-228.
- [9] Yamada, S., Omori, T., Ohnishi, A. (2019) "Verification method of reliability requirements", *Proceedings of the 23rd International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES) (Procedia Computer Science vol. 159)*: 860-869
- [10] Kaiya, H., Ohnishi, A. (2011) "Quality Requirements Analysis Using Requirements Frames", *11th International Conference on Quality Software*: 198-207
- [11] Dal Cin, M. (2000) "Structured language for specifications of quantitative requirements", *Proceedings of the 5th IEEE International Symposium on High Assurance Systems Engineering (HASE 2000)*: 221-227
- [12] Hayashi, K., Suzuki, A., Fujito, T., Fujimoto, K., Kobayashi, N. (2020) "A pre-handling method for reusing expert's knowledge on specification reviews", *JSAI SIG Technical Report, KSN-027-03*: 1-6, (in Japanese)
- [13] Roger S. Pressman, (2000) "Software Engineering: A Practitioner's Approach (5th Edition)", McGraw-Hill
- [14] Sugimoto, S., Minamida, K., Harada, Y. (2017) "Extracting Difference in Parenthetical Information from Software Documentations", *Proceedings of the Software Engineering Symposium 2017*: 265-269, (in Japanese)